

Program 6 Create a student management system using React. Implement routing to manage different views such as a student list and individual student profiles. Develop functionalities within the admin panel for adding, editing, and removing students, ensure seamless navigation between these views while displaying relevant information on each page.

Step 1: Create Project

```
npx create-vite@latest student-management --template react
cd student-management
npm install react-router-dom
```

Step 2: Create Folder Structure

```
src
├── components
│   ├── StudentList.jsx
│   ├── StudentProfile.jsx
│   └── StudentForm.jsx
├── App.jsx
├── main.jsx
└── index.css
```

Step 3: src/main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./index.css";

const root = ReactDOM.createRoot(document.getElementById("root"));

root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

Step 4: src/App.jsx

```
import React, { useState } from "react";
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";

import StudentList from "./components/StudentList";
import StudentProfile from "./components/StudentProfile";
import StudentForm from "./components/StudentForm";
```

```

function App() {
  const [students, setStudents] = useState([
    { id: 1, name: "John", age: 22, course: "CSE" },
    { id: 2, name: "Alice", age: 21, course: "IT" }
  ]);

  const addStudent = (student) => {
    setStudents([...students, { ...student, id: Date.now() }]);
  };

  const updateStudent = (updatedStudent) => {
    setStudents(
      students.map((s) =>
        s.id === updatedStudent.id ? updatedStudent : s
      )
    );
  };

  const deleteStudent = (id) => {
    setStudents(students.filter((s) => s.id !== id));
  };

  return (
    <Router>
      <Routes>

        <Route
          path="/"
          element={
            <StudentList
              students={students}
              deleteStudent={deleteStudent}
            />
          }
        />

        <Route
          path="/student/:id"
          element={<StudentProfile students={students} />}
        />

        <Route
          path="/add"
          element={<StudentForm addStudent={addStudent} />}
        />

        <Route
          path="/edit/:id"
          element={

```

```

        <StudentForm
          students={students}
          updateStudent={updateStudent}
        />
      }
    />

  </Routes>
</Router>
);
}

export default App;

```

Step 5: src/components/StudentList.jsx

```

import React from "react";
import { Link } from "react-router-dom";

function StudentList({ students, deleteStudent }) {
  return (
    <div className="container">

      <h2>Student Management System</h2>

      <Link to="/add">
        <button className="add-btn">
          Add Student
        </button>
      </Link>

      <ul>
        {students.map((s) => (
          <li key={s.id}>

            <Link to={` /student/${s.id} `}>
              {s.name}
            </Link>

            <button
              className="delete-btn"
              onClick={() => deleteStudent(s.id)}
            >
              Delete
            </button>

            <Link to={` /edit/${s.id} `}>
              <button className="edit-btn">
                Edit
              </button>
            </Link>
          </li>
        ))}
      </ul>
    </div>
  );
}

```

```

        </Link>

      </li>
    )}
  </ul>

</div>
);
}

export default StudentList;

```

Step 6: src/components/StudentProfile.jsx

```

import React from "react";
import { useParams, Link } from "react-router-dom";

function StudentProfile({ students }) {

  const { id } = useParams();

  const student = students.find(
    (s) => s.id === parseInt(id)
  );

  if (!student)
    return <h3>Student not found</h3>;

  return (
    <div className="container">

      <h2>Student Profile</h2>

      <p>
        <b>Name:</b> {student.name}
      </p>

      <p>
        <b>Age:</b> {student.age}
      </p>

      <p>
        <b>Course:</b> {student.course}
      </p>

      <Link to="/">
        <button className="submit-btn">
          Back
        </button>
      </Link>
    </div>
  );
}

```

```

    </div>
  );
}

export default StudentProfile;

```

Step 7: src/components/StudentForm.jsx

```

import React, { useState } from "react";
import { useNavigate, useParams } from "react-router-dom";

function StudentForm({
  addStudent,
  updateStudent,
  students = []
}) {

  const { id } = useParams();
  const navigate = useNavigate();

  const existingStudent = students.find(
    (s) => s.id === parseInt(id)
  );

  const [formData, setFormData] = useState(
    existingStudent || {
      name: "",
      age: "",
      course: ""
    }
  );

  const handleChange = (e) => {
    setFormData({
      ...formData,
      [e.target.name]: e.target.value
    });
  };

  const handleSubmit = (e) => {

    e.preventDefault();

    if (existingStudent) {
      updateStudent({
        ...formData,
        id: existingStudent.id
      });
    } else {

```

```
    addStudent(formData);
  }

  navigate("/");
};

return (
  <div className="container">

    <h2>
      {existingStudent ? "Edit Student" : "Add Student"}
    </h2>

    <form onSubmit={handleSubmit}>

      <input
        type="text"
        name="name"
        placeholder="Enter Name"
        value={formData.name}
        onChange={handleChange}
        required
      />

      <br /><br />

      <input
        type="number"
        name="age"
        placeholder="Enter Age"
        value={formData.age}
        onChange={handleChange}
        required
      />

      <br /><br />

      <input
        type="text"
        name="course"
        placeholder="Enter Course"
        value={formData.course}
        onChange={handleChange}
        required
      />

      <br /><br />

      <button
        type="submit"
```

```

        className="submit-btn"
      >
        Submit
      </button>

    </form>

  </div>
);
}

export default StudentForm;

```

Step 8: src/index.css

```

body {
  background-color: #f3f4f6;
  font-family: Arial, sans-serif;
}

.container {
  width: 60%;
  margin: 40px auto;
  background: white;
  padding: 25px;
  border-radius: 10px;
  box-shadow: 0px 3px 10px rgba(0,0,0,0.2);
}

h2 {
  text-align: center;
  color: #1e3a8a;
}

ul {
  list-style: none;
  padding: 0;
}

li {
  background: #eef2ff;
  padding: 15px;
  margin: 10px 0;
  border-radius: 8px;
}

a {
  text-decoration: none;
  color: #2563eb;
  font-weight: bold;
}

```

```
}

input {
  width: 95%;
  padding: 10px;
  border: 1px solid gray;
  border-radius: 5px;
}

button {
  padding: 8px 15px;
  margin-left: 10px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
}

.add-btn {
  background: green;
  color: white;
}

.edit-btn {
  background: orange;
  color: white;
}

.delete-btn {
  background: red;
  color: white;
}

.submit-btn {
  background: blue;
  color: white;
}
```

Step 9: Run

```
npm run dev
```

Open:

<http://localhost:5173>